Mootez Aloui

Cybersecurity-Focused Software Engineer — Security Specialist

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github.com/mootezaloui | Mootez Aloui | Portfolio

Fresh graduate specializing in the intersection of cybersecurity and artificial intelligence. Experienced in developing secure AI systems, vulnerability assessment for machine learning models, and implementing robust security measures in AI-driven applications. Published researcher in LLM security with hands-on experience in threat detection and adversarial machine learning.

Education

Mediterranean Institute Of Technology (MedTech) — Tunis, Tunisia
Software Engineering *— Cybersecurity & AI Specialization* — *Sept. 2020 – June 2025*

- **Capstone Focus:** Advanced research in LLM security vulnerabilities and adversarial attack

Research & Publications

PromptTrend Research — *Author & Researcher* — 2025

- **Published on arXiv:** *"PromptTrend: Continuous Community-Driven Vulnerability Discovery and Assessment for Large Language Models"*
- Pioneered novel approach to automated LLM vulnerability detection using community-driven threat intelligence
- Developed real-time monitoring system that identifies emerging prompt injection and adversarial attack vectors
- **Impact:** Framework adopted by security researchers for continuous LLM threat assessment

Professional Experience

Cybersecurity & AI Engineer Intern — Datadoit
LLM Security, Python, Docker, Cloud — *Jan. 2025 – June 2025*

- Designed and implemented **PromptTrend**, a continuous vulnerability discovery system for Large Language Models (LLMs).
- Published the research paper *"PromptTrend: Continuous Community-Driven Vulnerability Discovery and Assessment for Large Language Models"* on arXiv (2025).

Software Engineer Intern — Redstart
Python, TensorFlow, Cloud, React — *Summer 2024*

- Pentested a web application and verified the integrity of the data then deployed the application through the cloud.
- Implemented ML model for Email generation and TensorFlow.

Freelancer — multiple clients
Python, Scikit-Learn, Pandas, React, Express — *Summer 2023*

- Designed and deployed web applications for documents classification, cutting manual sorting time by 50%.
- Ensured secure integration of machine learning models into React web applications.

Junior Student Intern — Carrier
X3, Sage3000 — *Summer 2022*

- Supported system design improvements through data analysis and simulations.

Apple Developer & Tester — Apple
Swift, CoreML, Xcode — *Sept. 2021 – Present*

- Member in Beta iOS testing with machine learning integrations, enhancing user experience and security.
- Collaborated on testing AI-driven features, ensuring high standards of security and performance.

Example Projects

Equation Solver Chatbot | *Python, Docker, Vue, TensorFlow* — Jan. 2024 – Present

- Built a secure chatbot using NLP techniques and TensorFlow to solve mathematical problems.
- Used Docker for isolated deployments and ensured secure execution of computational tasks.

Skills

Programming Languages & Technologies: Python, TensorFlow, Keras, Docker, AWS, Azure, Linux, Bash, Git, Microservices Architecture, NLP, Data Analysis, Data Visualization, Model Deployment
Cybersecurity Skills: Vulnerability Assessment, Threat Detection, Secure Pipelines, OWASP Practices, Penetration Testing.
Development Practices: Agile/Scrum, TDD, BDD, Code Reviews, System Design, Security Best Practices, Secure Coding Standards
Certifications: Certified Scrum Master, AWS Cloud Practitioner, Google Cyber Security Professional Certificate, NVIDIA: Exploring Adversarial Machine Learning
Languages: French (native), English (professional), Arabic (native)

Awards & Recognition

Academic Excellence Awards 2022 – 2024

- **Best IoT Project (2022):** Innovative IoT project.
- **Best Web/Mobile Project (2023):** most innovative mobile and web applications for turtoring problems.
- **Best AI/ML Project (2024):** Groundbreaking work in adversarial machine learning defense

Research Impact 2025

- **Published Researcher:** First-author publication in LLM security on arXiv platform
- **Security Innovation:** Developed novel approach to continuous AI vulnerability discovery

Passionate about securing the future — Committed to ethical technology development — Ready to innovate in cybersecurity